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Microtubule-Associated Proteins and Microtubule-Interacting Proteins

Regulators of Microtubule Dynamics

*Maria Kavallaris, Sima Don,
and Nicole M. Verrills*

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SUMMARY

Microtubules are regulated by a range of proteins that interact with tubulin and regulate their stability. A large number and variety of microtubule-associated proteins (MAPs) and microtubule-interacting proteins have been identified and they exhibit cell and tissue specific expression. MAPs and microtubule-interacting proteins carry out a wide range of functions including regulation of microtubule stability, cross-linking microtubules and mediate interactions of microtubules with other proteins in the cell. The dynamic nature of microtubules and their range of cellular functions is dependent of the interaction and regulation of MAPs and microtubule-interacting proteins.

Key Words: Microtubules; tubulin; microtubule associated proteins; stathmin; microtubule stability.

1. INTRODUCTION

Microtubules perform a spectrum of functions and are a major component of the cytoskeleton. They are dynamic structures involved in many cellular processes including cell division, intracellular transport, and certain forms of cellular movement. The diversity of microtubule functions are regulated and influenced by microtubule-associated proteins (MAPs) and a range of tubulin and microtubule-interacting proteins. Indeed, the complex process that leads to the defined temporal and spatial organization of

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microtubules involves both structural and regulatory coordination of MAPs and associated proteins. Microtubules are polymers that consist of $\alpha\beta$ -tubulin heterodimers that are present in all eukaryotes. They are highly dynamic structures that are constantly growing and shortening, behavior referred to as dynamic instability. Multiple α - and β -tubulin isotypes have been described in mammalian cells and they display both developmental and tissue-specific expression (1). The various isotypes share a high degree of amino acid homology although they vary in their carboxy terminal region. It has been suggested that the role of different β -tubulin isotypes may be to influence microtubule dynamics by binding distinct MAPs in this region (2). Multiple factors can regulate the assembly and stability of microtubules including differential expression of different tubulin isotypes, regulation of monomer folding through tubulin-folding cofactors, microtubule nucleation, and post-translational modification of tubulin proteins. Additionally, various functions of microtubules involve their interaction with a large number of microtubule interacting proteins, which influence the regulation and distribution of microtubules in the cell through stabilizing or destabilizing effects. There is a wide variety of MAPs that provide the functional diversity of microtubules. This chapter will focus on MAPs that are primarily involved in regulating microtubule stability and dynamics.

2. STRUCTURAL AND FUNCTIONAL FEATURES OF MAPS

Structural MAPs were initially identified as proteins that copurified with tubulin in vitro, promoted microtubule assembly and stabilized microtubules. A variety of structural MAPs can interact on the surface of microtubules and reduce their dynamic behavior by stimulating tubulin polymerization. The precise nature of MAP binding to microtubules is not known. MAPs have a positive charge and it has been generally believed that they interact with the negatively charged C-terminal end of tubulin, facilitating the tubulin subunits to form the microtubule polymer. The exact sequences on tubulin that interact with MAPs are not well defined. This is partially because of the fact that little is known about the tertiary structure of the C-terminal region of tubulin.

Structural MAPs such as Tau, MAP2, and MAP4, are heat-stable and share a conserved molecular structure consisting of a projection domain and a carboxy-terminal microtubule-binding domain (3–8) (Fig. 1). This conserved domain contains three or four pseudorepeats. The N-terminal projection domain protrudes from the microtubule wall but does not bind microtubules, whereas the C-terminal microtubule binding domain (MTB) binds to the microtubule (9,10). The projection domain, projects away from the microtubule wall and is able to crosslink the microtubule with membranes, other microtubules or intermediate filaments (11).

MAPs colocalize with microtubules and MAP staining follows a similar structural pattern to microtubule staining (Fig. 2). As the predicted function of structural MAPs is to stabilize microtubules, there must be a mechanism that regulates these interactions in order for the dynamic feature of microtubules to exist. Phosphorylation of MAPs causes them to dissociate from microtubules, hence, allowing the microtubules to depolymerize leading to increased dynamic instability. The phosphorylation event occurs at the MTB of MAPs by several kinases such as the MAP kinases (12,13) and cdc2 kinase (14). Microtubule-affinity-regulating kinases phosphorylate MAPs at their MTB domain, causing their disengagement from the microtubule and leading to increased dynamic instability (15).

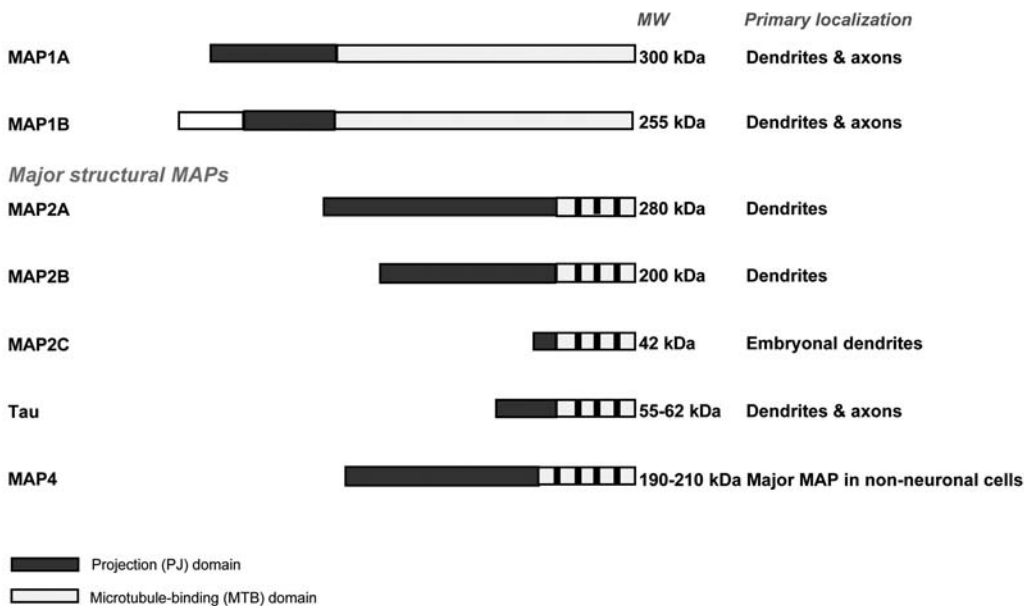


Fig. 1. Schematic diagram of the major MAPs. Protein structure, Protein size, and primary cellular localization. To view this figure in color, see the insert and the companion CD-ROM.

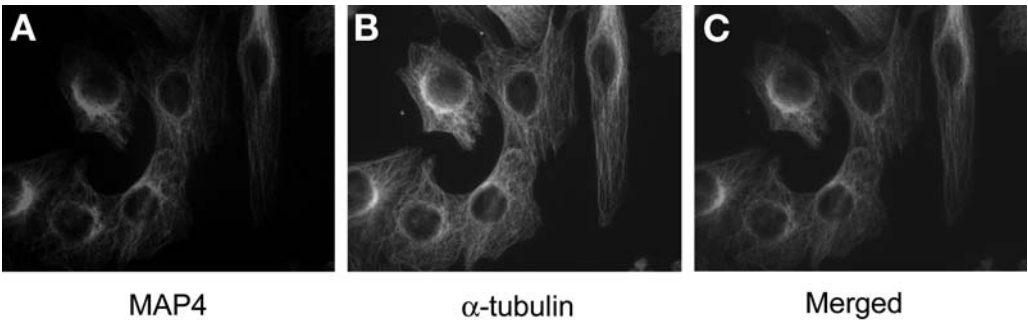


Fig. 2. MAP binding to microtubules. Human neuroblastoma, SHEP, cells costained with antibodies against MAP4 and α -tubulin and visualized using secondary antibodies tagged with Cy3 (red) and Cy2 (green), respectively. (A) MAP4 staining demonstrates that the MAP follows a microtubule-like distribution pattern. (B) α -Tubulin staining reveals the intricate microtubule network in these cells. (C) Merged image showing the costaining and localization of MAP4 and microtubules (yellow/orange). To view this figure in color, see the insert and the companion CD-ROM.

The organization of microtubules in different cell types is likely to be dependent on the types of MAPs expressed and the nature of the microtubule-organizing center. The basis for this is that the organization of microtubules varies in different cell types. As an example, interphase microtubules in cultured cells tend to radiate from the centrosome to the cell periphery (16) while microtubules in neurites run in parallel structures (17). The major structural MAPs in neurons are MAP2 and tau (12), whereas the most abundant and ubiquitous MAP in nonneuronal cells is MAP4. Many of the functional differences in the neuronal and nonneuronal MAPs have been identified in cells transfected with cDNA encoding MAP2, tau or MAP4. Distinct differences in the

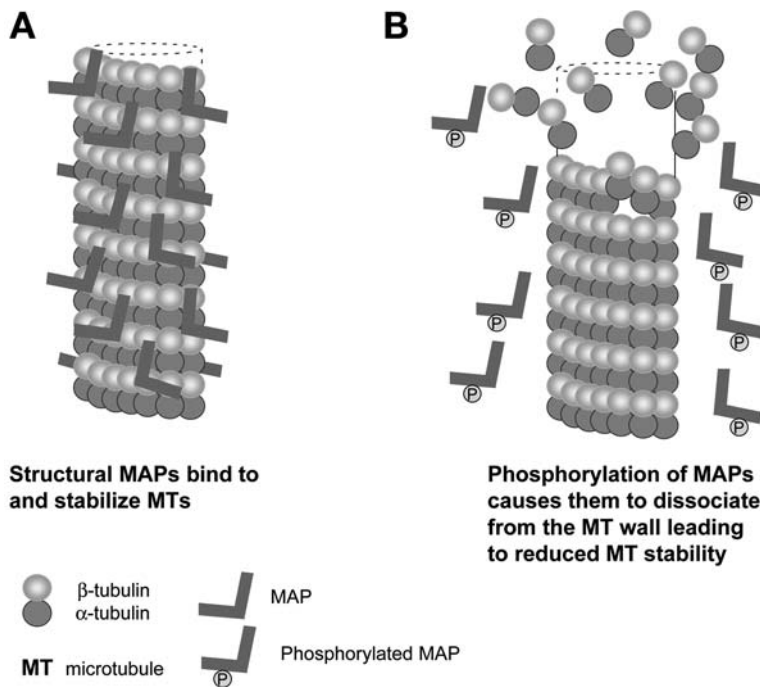


Fig. 3. Schematic model of interactions of MAPs with microtubules. **(A)** Structural MAPs such as MAP2, tau, and MAP4 can bind through their MBD to the microtubule wall and suppress microtubule dynamics by stabilizing microtubules. **(B)** The interaction of MAPs with microtubules is regulated by phosphorylation. Phosphorylation causes MAPs to dissociate from the microtubule wall, which in turn increases microtubule dynamics. NB: The size of the projection domain varies between the various MAPs (*see* Fig. 1) and therefore this figure is a generalized representation of MAP-microtubule interactions. To view this figure in color, see the insert and the companion CD-ROM.

organization of microtubules have been noted in cells transfected with cDNA encoding these three MAPs. **Dense microtubule bundling is induced with MAP2 or Tau (18–20) whereas this does not occur with MAP4 (21,22). MAPs are differentially expressed in different tissues** and are critical regulators of microtubule assembly and stability (Fig. 3).

2.1. MAP1

The microtubule-associated proteins 1A (MAP1A) and 1B (MAP1B) are high molecular weight protein complexes that are distantly related and are encoded by distinct genes that reside on chromosomes 15q15.3 and 5q13, respectively. Both MAP1A and MAP1B are synthesized as a polypeptide (referred to as the heavy chain) coupled to a light chain (LC), i.e., MAP1A-LC2, 28 kDa, and MAP1B-LC1, 34 kDa (23,24). MAP1 proteins are thought to play a role in regulating the neuronal cytoskeleton. MAP1A is abundant in the adult brain, whereas MAP1B is a neuritegenesis-associated MAP (25). MAP1A binds to microtubules through a novel acidic binding motif (26), whereas MAP1B binds to microtubules through a series of basic motifs (27). A third light chain, LC3 (18 kDa), is a subunit of MAP1A and MAP1B and has been proposed that expression of LC3 can regulate the microtubule binding activity of MAP1A and MAP1B (28).

MAP1B is the first MAP to be expressed during embryonal development of the nervous system. The product of the MAP1B gene is a precursor polypeptide that most likely undergoes proteolytic processing to generate the final MAP1B heavy chain and LC1 light chain (reviewed in ref. 29). In contrast to other microtubule-stabilizing proteins such as MAP2 and tau that differ in their molecular structure (Fig. 1), transfection of MAP1B into COS cells did not result in the formation of microtubule bundles (30). However, microtubules in MAP1B-transfected cells were stabilized against nocodazole-induced depolymerization and were enriched in acetylated α -tubulin (indicative of stable microtubules) (30). Therefore, MAP1B can stabilize microtubules in cells, albeit less efficiently than MAP2 and tau, but it cannot induce microtubule bundling.

In mice, MAP1B is expressed at high levels during embryogenesis and reaches maximum levels at 2–3 wk of age followed by a decrease in MAP1B and increased expression of other MAPs such as MAP1A. The highest levels of MAP1B are found in regions that show extensive growth of axons and motor neurons, retinal ganglion cells, olfactory epithelium, and nerve layer of olfactory bulb (31–33). The distribution and function of MAP1B can be modified by phosphorylation. Studies on neuronal growth cones have revealed that MAP1B can interact with both microtubules and actin microfilaments (34,35). Gene knockout studies of the murine MAP1B gene suggested an important role in development and function of the nervous system (36,37).

2.2. Tau

Brain tissue is an abundant source of microtubules and many studies have traditionally focused on brain MAPs. The most studied MAPs are the heat stable proteins MAP2 and Tau. Both these neuronal MAPs share homologous repeats in the carboxy-terminal region that contribute to microtubule binding and stability (38). Tau has been shown by covalent crosslinking experiments to bind to two distinct sites on the C-terminal third of both α - and β -tubulin (39).

The Tau gene is located on chromosome 17q21. Tau tissue expression is predominantly neuronal, although very low levels are detected in oligodendrocytes and astrocytes. Tau mRNA and proteins have also been detected in diverse tissues of nonneuronal origin such as heart, kidney, lung, muscle, pancreas, and testis (40–42). The 5' UTR to the end of the 3' UTR of Tau, spans 133.9kb and contains 16 exons (41,43). Although exons 1 and 14 are transcribed, they are not translated. Tau exons are alternatively spliced in a tissue-specific and a developmentally regulated manner. Certain isoforms of Tau mRNA such as 4A, 6, and 8 are never found in brain. However, 4A is found in peripheral nervous tissue and results in a tau protein isoform originally named big tau for its higher molecular weight (44).

Transcription of Tau starts at a unique site at the start of exon 1 and stops at one of two alternate polyadenylation sites downstream of exon 14. This leads to two transcripts of 2 and 6 kb encoding tau, with the 6 kb being the major isoform in brain. Interestingly the two transcripts differ in their subcellular localization, with the 2 kb isoform found in the cell body and the 6 kb isoform found in axons (45). Tau gene regulation is poorly understood. It is thought that the 3' UTR contains *cis*-elements that localize tau mRNA to specific cell compartments. The half-life of Tau mRNA is relatively long in neuronal cells (~10 h).

There are three adult-brain-specific alternatively spliced coding exons, exons 2, 3, and 10. Incorporation of exon 10 adds one microtubule-binding repeat domain to the

three other exons that alter the tau protein binding capacity on microtubules. Tau exists as two classes of protein because of alternate splicing of exon 10, 3 repeat (3R)- and 4 repeat (4R)-tau isoforms (46). Further, alternate splicing of exons 2 and 3 allows for a total of six isoforms, three each of 3R and 4R isoforms. In the foetal brain only the shortest isoform is expressed (46).

Tau is localized mainly in axons, which has several protein isoforms ranging in size from 55 to 62 kDa (47). Tau proteins were shown to be potent promoters of tubulin polymerization (48), and their overexpression by microinjection and transfection led to increased microtubule mass accumulation, increased microtubule stability, and extensive microtubule bundling (18,49). Inhibition of Tau isoforms by antisense oligonucleotides prevented the establishment of axonal polarity and outgrowth in primary cerebellar neurons (50). Furthermore, sense and antisense transfection analysis of Tau function in PC 12 cells demonstrated the role of tau in promoting net microtubule assembly, the rate of neurite elongation and stability of neurites (51). In contrast to the in vitro studies outlined above, mice with targeted disruption of the Tau gene exhibited subtle changes in microtubule organization in small caliber axons and despite being chronically deficient in tau, were relatively normal (52). Interestingly, the total absence of this usually predominant MAP did not alter microtubule stability or neurite growth properties in primary neuronal cultures from mice. A possible explanation for the stark differences between the in vitro and in vivo studies is that there is sufficient MAP protein redundancy or adequate plasticity to mask the requirement for tau during development in the mouse. Gene targeting of another predominant axonal MAP, MAP1B also produced a viable and relatively normal mouse (37). However, mating *tau* and *MAP1B* null animals demonstrated that loss of both axonal MAPs is lethal by 4 wk of age (53). It appears that both *tau* and MAP1B are required for the genesis of axon tracts and that these two MAPs function synergistically.

There has been a strong association between the accumulation of filamentous tau and a number of neurodegenerative diseases, referred to as tauopathies. These include, Pick's disease, frontal temporal dementia, and Parkinsonism linked to chromosome 17 (FTDP-17), and Alzheimer's disease (54). All these conditions lead to major filamentous deposits of hyperphosphorylated tau with associated neuronal loss in the affected regions. Alzheimer's disease is the most common human cognitive neurodegenerative disease. The principle pathology in the brain neurons of affected individuals are insoluble tangled filaments, which are predominantly made up of tau aggregations (55). The identification of tau mutations in a set of dominantly inherited cases of FTDP-17 was a major breakthrough in unraveling neuronal degradation and death. A direct link between errors in tau and abnormal accumulation of tau-containing filaments in neurons, have been directly associated with neuronal failure (reviewed in ref. 54). Two classes of tau mutations have been identified that lead to morphologically distinct filaments. One class of mutations can alter the binding of tau to microtubules and leads to the production of abnormal twisted filamentous structures that are made up of all six isoforms that occur primarily in neurons. The other class is splicing mutations that change the ratio of tau with four microtubule-binding domains (4R) to types with only three microtubule-binding domains. These mutations cause the appearance of filamentous structures in both neurons and glia. Despite the differential effects of the two classes of mutations, they both lead to FTDP-17, possibly because they cause a net increase in the unbound pool of tau. Increased free tau in neurons could lead to sequestration of free tubulin as was observed

in lamprey anterior bulbar cells overexpressing human tau (56). These cells contained tau filaments entangled with tubulin.

2.3. MAP2

MAP2 is a neuronal-specific high molecular weight protein ($M_r \sim 280$ kDa) (57), which is able to bind both tubulin and actin through its tubulin-binding domain and modify the stability of microtubules and microfilaments (58,59). Both MAP2-tubulin and MAP2-actin interactions are regulated by phosphorylation (59), however, the phosphorylation events that reduce the ability of MAP2 to bind to the microtubules enhance its ability to interact with the actin cytoskeleton (60). The MAP2 gene is located on chromosome 2q34-q35. The expression of MAP2 is predominantly neuronal and more specifically, in dendrites. Multiple isoforms of MAP2 exist in human cells, with high molecular weight isoforms MAP2a and MAP2b, and low molecular weight isoforms MAP2c and MAP2d being expressed at different stages of development (61) and differentiation (62). MAP2a and MAP2b are concentrated in dendrites of neuronal cells, with MAP2a being expressed at the late stage of development and MAP2b being present at both embryonic and adult stages. MAP2c is also expressed in dendrites, but can be found in axons and glial cells (25). Different isoforms of MAP2 stabilize microtubules to different extents. Although MAP2c is usually expressed at the early stages of development and then replaced by high molecular weight MAP2 isoforms (61), it has been shown to be expressed continuously in the adult mammalian olfactory system and in the retina (33). MAP2c is expressed at high levels in early brain development. MAP2a and MAP2b localize exclusively to dendrites, whereas MAP2c is found in all cell compartments (63,64), suggesting that the various MAP2 isoforms have distinct cellular functions. MAP2d is produced by developmentally regulated alternate splicing of MAP2 (65,66).

Apart from regulation of cytoskeletal dynamics, other functions of MAP2 include neuronal morphogenesis, where it has been tentatively implicated in neuronal outgrowth and polarity, microtubule crosslinking in dendrites as well as organelle trafficking in axons and dendrites (25,67). Inhibition of MAP2 expression by antisense transfection led to inhibition of processes associated with neuronal differentiation such as formation and outgrowth of neurites (68). Moreover, in MAP2-deficient mice, a reduction in microtubule density in dendrites and a reduction of dendritic length were observed, demonstrating that this protein is essential for dendrite formation and growth (69). MAP2c is also involved in the neurite initiation process, both by stabilizing microtubules and re-organizing filamentous actin structures in forming neurites (70).

MAPs such as MAP2, have been associated with changes in microtubule stability in antimicrotubule drug resistant cells. Vincristine-resistant neuroblastoma cells, BE/VCR10 that display increased levels of polymerized tubulin, were investigated for changes in MAP expression (71). A marked decrease in MAP2c was identified in the vincristine-selected cells compared with the drug-sensitive parental, BE(2)-C cells. In contrast, no major changes were observed in MAP2a, MAP2b, or MAP4 expression in the resistant cells. Several studies have suggested that MAP2c promotes microtubule polymerization to a much lesser extent than high molecular weight MAPs (72,73). Moreover, the significant decrease in expression of MAP2c in BE/VCR10 cells is likely to be contributing to hyperstable microtubules by shifting the equilibrium toward MAP2 isoforms such as MAP2a and MAP2b that are stronger promoters of microtubule stability than MAP2c. This in turn would counteract the microtubule

destabilizing effects of vincristine, thus leading to a selective growth advantage of these cells in the presence of the drug.

2.4. MAP4

The majority of studies on structural MAPs have focused on those originally identified in neuronal tissues and cells. The major MAP found in cells of non-neuronal origin is MAP4. The Map4 gene resides on chromosome 3p21, and encodes for a heterogeneous thermostable polypeptide with a molecular weight of 190–210 kDa (Fig. 1). The heterogenous nature of the MAP4 protein may be as a result of alternative splicing during transcription (74) and/or post-translation phosphorylation (75). It was first isolated from human HeLa cells as a 210 kDa protein (76) the same size as that found in mouse tissues (77). However, in rat and bovine species, the molecular weights are 200 kDa and 190 kDa, respectively as identified on sodium dodecyl sulfate-polyacrylamide gel electrophoresis (78). Homologs of MAP4 have been identified in all vertebrates and even nematodes (79). Microtubules with wide ranging functions are associated with MAP4 including those involved in organelle and vesicular transport, cell shape during differentiation and spindle microtubules during mitosis (16,80).

Overexpression of either full-length MAP4 or the microtubule-binding domain of MAP4 has been shown to stabilize microtubules in stably transfected mouse L^{tk} cells that either contained the full-length MAP4 (L-MAP4) or its L-MTB (22). More stable microtubules were observed in both transfected cell lines, as determined by exposure to nocodazole, an agent that depolymerizes microtubules (22). In addition to microtubule stability, cell growth was affected. Of interest was a study that aimed to see if this effect on cell growth influenced the *in vivo* function of microtubules. Overexpressing MAP4 was found to inhibit vesicular transport such as recycling of the transferrin receptor, low-density lipoprotein (LDL), and some Golgi elements in addition to stabilizing microtubules (81). This finding added support for the importance of MAP4 in regulating microtubule function.

Similarly, biochemical studies have demonstrated the *in vitro* ability of MAP4 to stabilize microtubules and dampen dynamic instability (76,82), and these findings are supported *in vivo* by transfection (21,22) and microinjection studies (83). Antisense RNA directed toward MAP4 caused a decrease in total tubulin, a decrease in polymerized tubulin, slow recovery from drug-induced microtubule depolymerization, and a less polar and flattened cell shape (84). This suggests that MAP4 regulates the assembly level of microtubules and through this mechanism may be involved in controlling cell spreading and shape. In addition, post-translational modifications of MAPs have been shown to regulate microtubule dynamics. For example, phosphorylation of MAP4 by cyclin B-cdc2 kinase renders microtubules more dynamic *in vitro* (85) and reduces MAP4's capacity to stimulate *in vitro* polymerization of microtubules (86). A nonphosphorylatable mutant of MAP4 was shown to bind microtubules more avidly than wild-type MAP4 *in vitro*, and microtubules in cells expressing nonphosphorylatable MAP4 were more resistant to nocodazole depolymerization (87).

MAP4 can localize both to interphase (Fig. 2) and mitotic microtubules (88), which suggests that it may contribute to the role of microtubules in proliferation and differentiation of cells. MAP4 has the same structural organization as the other members of the MAP family of proteins, i.e., it has an acidic domain, which projects from the surface of the microtubule and a basic domain that interacts with the microtubule wall (21). The

assembly promoting portion of the microtubule-binding domain has a sequence that is responsible for binding to the microtubules (89). This indicates that MAP4 interacts directly with microtubules and that altering this interaction can influence microtubule dynamics and assembly.

In the mouse, expression of MAP4 was found in most tissues. MAP4 was predominant in specific cell types indicating that it may play a functional role in particular tissues or cells (80). **Multiple isoforms of MAP4 appear to be tissue and developmental stage specific (88).** A detailed analysis of MAP4 microtubule-binding domains has revealed that the isoforms are distinct from each other (88). Together, these data support the view that the different MAP4 isoforms may have differing roles in specific tissues.

Polymerized microtubules exist in equilibrium with free tubulin dimers. The main function of MAP4 is to stabilize microtubules, thus increased expression of MAP4 may shift this equilibrium toward more stable microtubules leading to decreased levels of free tubulin dimers. An antisense study demonstrated that depleting MAP4 expression reduced the level of cellular tubulin and also decreased the stability of microtubules (84). These contradicted previous results, which showed that the absence of MAP4 had no effect on microtubule stability (90). It should be noted that these studies differed in their approach. The initial MAP4 functional study involved microinjection of antibody that blocked MAP4 binding to microtubules into single cells and studied the effect on microtubule dynamics (90). This approach did limit the studies that could be performed because of the low cell number. In contrast, the other group stably expressed antisense MAP4 in cells, which allowed a more detailed analysis of microtubule dynamics, morphology and cell cycle to be performed (84).

Changes in MAP4 expression have been associated with resistance to antimicrotubule agents. In leukemia cells selected for resistance to the microtubule destabilizing agents vincristine, a significant increase in MAP4 protein expression was detected (91). The vincristine-resistant cells had an acquired mutation in β -tubulin and increased polymerized tubulin levels. The increased MAP4 protein was found to be associated with the microtubule fraction, suggesting it was not phosphorylated, and therefore contributing to the increased microtubule stability in these cells.

In murine cells, induction of the tumor suppressor gene, p53 transcriptionally represses MAP4 (92). **Increased MAP4 expression, which occurs when p53 is mutated, increases microtubule polymerization and paclitaxel binding, resulting in heightened sensitivity to the microtubule stabilizing drug paclitaxel and reduced sensitivity to the microtubule destabilizing drug, vinblastine (93).** Thus, controlling this p53-dependent regulation of MAP4 is a potential approach to enhance the action of antimicrotubule drugs. DNA damage increased wild-type p53 and decreased MAP4 expression, resulting in decreased sensitivity to paclitaxel and increased sensitivity to vinblastine in cell lines (94). This regulation is now being tested in cancer clinical trials where the DNA damaging agent, doxorubicin is used to induce p53, and hence repress MAP4, followed by sequential treatment with vinorelbine (95).

3. OTHER REGULATORS OF MICROTUBULE STABILITY

The focus in the preceding section was on MAPs that bind to and stabilize microtubules. There are however, other regulators of microtubule function that affect the dynamic properties of microtubules by affecting the stability of microtubules. Microtubule-destabilizing

proteins include a group of proteins that upon binding to the microtubules or the tubulin heterodimers, decrease the total microtubule polymer mass. Examples of such proteins include stathmin, mitotic centromere-associated kinesin (MCAK), and katanin and examples of microtubule-stabilizing proteins include stable tubulin only protein (STOP), End-binding (EB) 1, CLIP-170, tumor overexpressed gene (TOG), and surviving. Many of these microtubule dynamic regulators play a pivotal role in cell division.

3.1. Other Microtubule Stabilizing Proteins

3.1.1. STABLE TUBULIN ONLY PROTEIN (STOP)

Another group of MAPs that promote microtubule stabilization are STOP proteins. These calmodulin-regulated and calmodulin-binding proteins are encoded by a single gene that displays highly variable tissue specificity because of mRNA splicing and alternate promotor use. The STOP gene is localized to chromosome 13q3. An interesting feature of these proteins is that, unlike other MAPs such as MAP2 and tau, they stabilize microtubule polymers against cold induced depolymerization (96). STOPS are expressed in both neuronal and nonneuronal cell types, and a lack of these proteins is thought to be associated with defects in synapse function in mice (97). STOP proteins are also involved in neuronal differentiation and are required for normal neurite formation (98).

3.1.2. END-BINDING PROTEIN 1 (EB1) AND OTHER PLUS END TRACKING PROTEINS

EB1 belongs to an evolutionary conserved family that regulates microtubule assembly and stability. EB1 was originally identified by its physical association with the carboxy-terminal portion of the adenomatous polyposis coli (APC) tumor suppressor protein, an APC domain commonly mutated in familial and sporadic forms of colorectal neoplasia (99). EB1 associates with APC and p150^{glued} (also known as dynactin 1), a component of the dynactin complex. EB1 colocalizes with microtubules, preferentially at their plus ends, throughout the cell cycle (100). Another term for the behavior of proteins that associate on microtubule ends is “plus-end tracking” proteins (+TIP) (101,102).

Overexpression of EB1 induces microtubule bundles that are more resistant to nocodazole and more acetylated than regular microtubules (100). Analysis of the functional domains of EB1 protein family members revealed overlapping but distinct regions that contain the C-terminal signature that associates with APC and p150^{glued} (103). Interestingly, the APC or p150^{glued} binding domains are not required for EB1- or the EBF3-induced microtubule bundling. EB1 is a potent inducer of microtubule polymerization and is a specific marker of growing microtubule tips (104–109). Apart from the interaction of EB1 with microtubule tips, both EB1 and APC localize to centrosomes and are functional components of this system (110). EB1 plays an important role in anchoring the minus end of microtubules to the centriole. EB1 binding also reflects kinetochore directionality during cell division suggesting that EB1 may have a functional role in the polymerization of kinetochore microtubules and/or attachment (111).

In recent years the number of identified +TIPs and their proposed mechanisms of action have been increasing (reviewed in ref. 112). The first evidence for the existence of +TIP was observed with live cell imaging between a fusion protein of a MAP called cytoplasmic linker protein, CLIP-170, and green fluorescent protein (113). CLIP-170 is a nucleotide-sensitive MAP that links microtubules to endocytic vesicles in vitro (114,115). The plus end tracking of CLIP-170 is mediated by microtubule treadmilling

(116). CLIP-170 proteins add to the plus ends of growing microtubules, but then soon these molecules dissociate behind a region of new microtubule growth giving the illusion that they are moving along with the microtubule although individual CLIP-170 molecules are stationary. All CLIP-170 family members have the ability to plus-end track and these proteins contain one or more conserved MTBs, referred to as CAP-Gly at their N-terminal end. This domain is also found in the p150^{glued} subunit of dynactin (a complex that regulates cytoplasmic dynein, a motor protein), certain tubulin folding factors and a member of the kinesin motor superfamily (117).

CLIP-170 and CLIP-115 associate specifically with the ends of growing microtubules as plus-end tracking proteins and may act as microtubule stabilizing factors. Two CLIP-associated proteins (CLASPs) have been identified (118). CLASPs have a number of functional features such as:

1. Ability to bind CLIPs and microtubules;
2. Colocalize with the CLIPs at microtubule plus ends; and
3. Exhibit microtubule-stabilizing effects in transfected cells.

CLASP2 appears to be involved in orientating stabilized microtubules toward the cells leading edge (118). Evidence suggests that CLIP-170 may mediate the association of dynein/dynactin to microtubule plus ends, and it binds to kinetochores in a dynein/dynactin-dependent fashion, in both cases, through its C-terminal domain. The C-terminal domain contains two zinc finger motifs that are thought to mediate protein-protein interactions. A protein implicated in normal brain development, LIS1 was originally identified in a rare brain malformation called Lissencephaly I, and it interacts with dynein and other proteins. LIS1 has been shown to colocalize and directly interact with CLIP-170. The recruitment of LIS1 to kinetochores is a dynein/dynactin dependent process, and CLIP-170 is dependent on its binding to LIS1 (119). Overexpression of CLIP-170 leads to localization of phospho-LIS1 and dynactin to stabilized microtubule bundles. LIS1 appears to play a role as a regulated adapter between CLIP-170 and cytoplasmic dynein at sites involved in microtubule transport, and/or in the control of microtubule dynamics.

3.1.3. Dis1/TOG

In recent years, a new family of MAPs, the human analog of TOG (Dis1/TOG) family has been identified as regulators of microtubule function (120). The Dis1/TOG family is highly conserved in evolution. Unlike the structural MAPs, the localization and function of Dis1/TOG proteins are not dependent on their microtubule-binding activity and they tend to perform their diverse roles by interacting with other regulatory molecules such as microtubule motors and centrosomal proteins. A 6,449 bp cDNA, termed colonic, hepatic tumor over-expressed gene (ch-TOG) that is highly expressed in human tumors and brain has been described and it encodes for a 218 kDa TOG polypeptide. The distribution of TOG is cell cycle-dependent and is associated with centrosomes and spindles in mitotic cells (121). In contrast, during interphase it concentrates in the perinuclear cytoplasm. TOG cosediments with paclitaxel-stabilized microtubules and TOG can promote microtubule assembly *in vitro* (121). TOG displays high homology to XMAP215, a previously described MAP from *Xenopus* eggs, and appears to be important for microtubule rearrangements and spindle assembly in dividing cells (120).

3.1.4. SURVIVIN

Survivin is a member of the inhibitor of apoptosis gene family and functions both as a suppressor of apoptotic cell death and as a regulator of cell division. It is localized to both kinetochores and mitotic spindle microtubules, and increases microtubule stability by either directly regulating microtubule dynamics or by recruitment of MAPs and motor proteins (122,123). Wild-type p53 negatively regulates survivin expression at the gene and protein levels, by repressing the survivin promoter (124). The role of survivin in the inhibition of apoptosis and in the regulation of microtubule dynamics may potentially facilitate evasion from checkpoint mechanisms of cell cycle arrest and promote resistance to antimicrotubule agents (122).

3.2. Microtubule Destabilizing Proteins

3.2.1. STATHMIN

A major tubulin-regulatory protein is stathmin (also referred to as oncoprotein 18, OP18; metastastin; p19), which was first identified as a highly overexpressed protein in leukemia (125,126) and as a protein that underwent phosphorylation in response to extracellular stimuli (127). Stathmin is a well-conserved, ubiquitous, cytosolic phosphoprotein that destabilizes microtubules and forms specific complexes with tubulin dimers (128,129).

The stathmin gene resides on chromosome 1q36, and expression has been detected in all tissues, with the highest levels found in brain, neurons, testis, and leukemic lymphocytes (130). Stathmin belongs to a family of structurally related proteins that are also expressed in the nervous system and include superior cervical ganglion (SCG)10, SCG10-like protein (SCLIP), RB3 and two splice variants RB3' and RB3'' that displays microtubule destabilizing activity (131,132) (Fig. 4). Stathmin interacts directly with microtubules (129), through a complex of one stathmin molecule to two tubulin heterodimers (133,134). Stathmin binds to the region around helix 10 of α -tubulin dimer, a region involved in longitudinal interactions in the microtubule, sequestering the dimer and linking two α/β -tubulin heterodimers (135). Insight into the interaction of stathmin with tubulin has been gained with the aid of a 3.5 Å model of tubulin in complex with colchicine and with the stathmin-like domain of RB3 (136,137). The RB3-stathmin-like domain resembles a hook-like appearance that holds two bound tubulin heterodimers (Fig. 5).

The interaction of stathmin with tubulin is directly dependent on the degree of stathmin phosphorylation, whereby increasing phosphorylation inhibits binding (138,139). The sequential cell cycle-dependent phosphorylation on four serine residues, Ser16, 25, 38, and 63 abolishes the microtubule-destabilizing activity of stathmin, with phosphorylation of Ser63 contributing substantially to the inactivation (reviewed in ref. 140). Two possible mechanisms of microtubule destabilization by stathmin have been proposed:

1. Sequestration of tubulin heterodimers resulting in depletion of the soluble tubulin available for polymerization (reviewed in ref. 141);
2. Stimulation of microtubule catastrophe (129).

It appears that both mechanisms can occur depending on pH, with tubulin-sequestering and catastrophe-enhancing activity observed at pH 6.8, and only catastrophe-enhancing at pH 7.5 (142). Decreased stathmin expression favors increased microtubule polymerization (143) whereas increased expression reduces microtubule polymer mass (138,144).

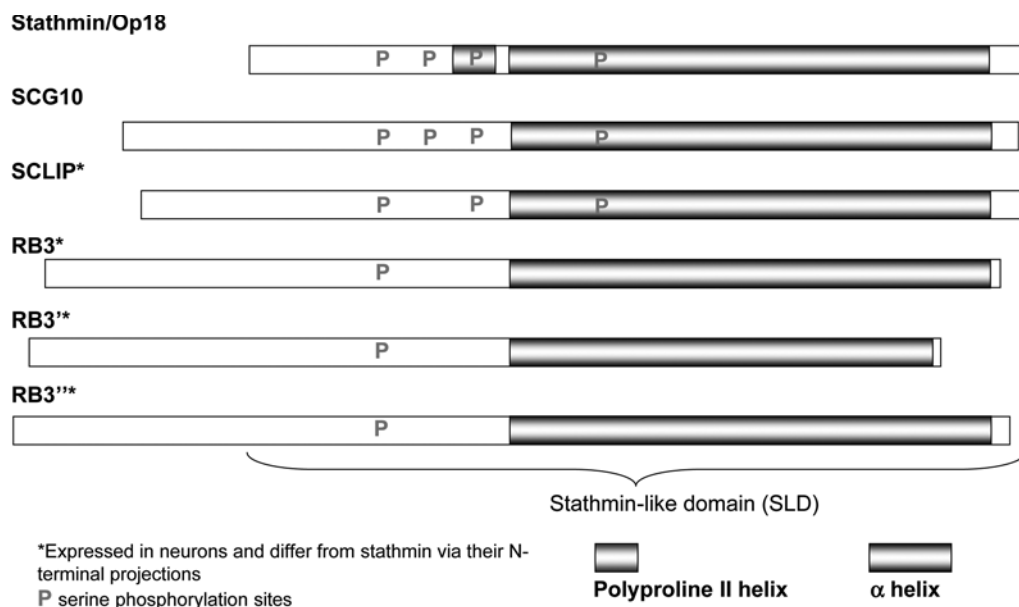


Fig. 4. Schematic diagram of the stathmin-family of proteins. The stathmin-like domain is highly conserved among the family members. Stathmin is unique among the family members as it contains a polyproline II helix in the N-terminal region. Apart from the remaining N-terminal region, which is not structured, the remaining protein has a α -helical structure. Adapted from ref. 168. To view this figure in color, see the insert and the companion CD-ROM.

Interestingly, increased stathmin levels have been identified in acute leukemia (125,145), lymphoma (145,146), neuroblastoma (147), and breast cancer (148) compared with normal tissue. Levels of stathmin phosphorylation significantly correlated with a high percentage of leukemic blast cells in S-phase and white blood count at diagnosis, suggesting that this could be a valuable target to inhibit the proliferation of leukemia cells (149). A somatic mutation in stathmin was recently identified in an esophageal adenocarcinoma (150). Transfection studies of the mutant stathmin into NIH3T3 cells resulted in foci formation and tumor growth when transplanted in immunodeficient mice. Cells expressing the mutant stathmin had altered tubulin ultra structure, and cell cycle analysis revealed a doubling in the percentage of cells in G2/M. The mutant stathmin had decreased specific phosphorylation, suggesting that mutations in stathmin affecting the phosphorylation capacity have profound effects on cell homeostasis that may lead to tumorigenicity (150). These studies suggest that regulation of microtubule dynamics may have a causal relationship to cancer development.

Cell lines displaying overexpression of stathmin show decreased microtubule polymerization, decreased paclitaxel binding, and decreased paclitaxel sensitivity (151). In paclitaxel-resistant A549 lung cancer cells with an α -tubulin mutation (152) and increased microtubule dynamic instability (153), the active nonphosphorylated form of stathmin was increased about twofold, whereas the inactive phosphorylated forms were barely detected (152). The two-dimensional polyacrylamide gel electrophoresis analysis of the childhood tumor neuroblastoma found that more aggressive neuroblastomas that coincidentally are less responsive to therapy, have reduced phosphorylation of stathmin (147).

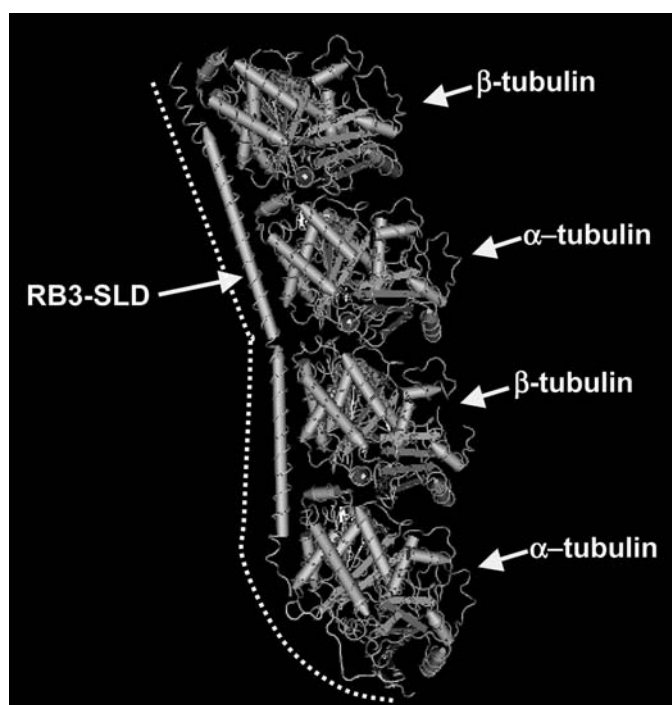


Fig. 5. Structure of tubulin-colchicine:RB3-tubulin complex. Insight into the interaction of stathmin with tubulin has been gained with the aid of a 3.5 Å model of tubulin in complex with colchicine and with the stathmin-like domain of RB3. The RB3-stathmin-like domain resembles a hook that holds two bound tubulin heterodimers (shadowed by the white dotted line for clarity) (Protein Data Bank entry 1SA1) (137). To view this figure in color, see the insert and the companion CD-ROM.

Stathmin, like MAP4 is negatively regulated by wild-type p53 (154,155). Antisense inhibition of stathmin was found to be synergistic with paclitaxel in inhibiting growth and clonogenic potential of K562 leukaemia cells (156). In combination with stathmin inhibition, paclitaxel induced more severe mitotic abnormalities and increased apoptotic effects. Inhibition of stathmin expression also resulted in increased sensitivity to paclitaxel and decreased sensitivity to vinblastine (157). In support, overexpression of stathmin in human lung carcinoma cells led to the increased sensitivity to *vinca* alkaloids vindesine and vincristine, but no changes in sensitivity to taxanes were observed (158). In human breast cancer cell lines that harbored mutations in p53 overexpression of stathmin was associated with decreased levels of microtubule polymer, and consequently, with decreased binding of paclitaxel and increased binding of vinblastine. Interestingly, stathmin-overexpressing cells were less sensitive to both of these antimicrotubule agents (151). Although the resistance of stathmin-overexpressing cells to paclitaxel was predicted according to altered microtubule dynamics and drug binding, resistance to vinblastine was unexpected, as the increased binding of the drug to the cells was observed. However, by measuring the amount of mitotic-specific phosphoproteins in controls and stathmin-overexpressing cells treated with vinblastine, it was found that stathmin decreases the number of cells entering mitosis. This impediment in cell progression from G₂ to mitosis was thought to be responsible for diminishing the cytotoxic effects of vinblastine (151).

3.2.2. KATANINS

Other microtubule-destabilizing proteins include *Xenopus* Kinesin Catastrophe Modulator 1 (XKCM1), *Xenopus* Kinesin Superfamily protein (XKIF)2 and katanin. XKCM1 and XKIF2 are Kin I kinesins that unlike most kinesins that use ATP to move along a microtubule, depolymerize microtubules. Katanin is a heterodimeric microtubule-severing protein that localizes to centrosomes (159) and is responsible for the majority of M-phase severing activity in *Xenopus* eggs (160), and is essential for releasing microtubules from neuronal chromosomes. (161). The kinesin-like proteins XKCM1 and XKIF2 are required for the formation and maintenance of the mitotic spindle. These proteins cause rapid microtubule disassembly in vitro, binding preferentially to microtubule ends, both plus and minus, and cause destabilization of the microtubule lattice (162,163). However, unlike stathmin, XKCM1 does not seem to possess tubulin-sequestering activity (reviewed in ref. 164).

3.2.3. MITOTIC CENTROMERE-ASSOCIATED KINESIN

In the mitotic spindle, kinetochore microtubules facilitate the separation and segregation of chromosomes (123). The functions of kinetochores include the attachment of the sister chromatids to the opposing poles of the spindle, generating force for the poleward chromosome movement, positioning chromosomes on the spindle plane, and control of anaphase/metaphase transition. Several motor and accessory proteins such as kinesins and dyneins, centromere proteins (CENP-B, -C, -E and -F), inner centromere protein (INCENP), and MCAK are involved in performing these functions (123,165). MCAK belongs to a unique group of motor proteins that are not motile but can destabilize microtubules. It is a member of the Kin I subfamily of kinesin-related proteins and it is an ATPase that catalytically depolymerizes microtubules by increasing the rate of dissociation of tubulin from the ends of microtubules (166).

The correct attachment of kinetochore microtubules is essential for the correct separation and segregation of chromosomes. The serine/threonine kinase Aurora B belongs to a family of proteins that constitutes key regulators in the execution of mitotic events. Aurora B is thought to regulate kinetochore-microtubule attachments and promote correct chromosome biorientation. MCAK can depolymerize microtubules. Both Aurora B and MCAK proteins have comparable functions where they localize to mitotic centromeres. They regulate microtubule dynamics, correct chromosome congression, and correct erroneous kinetochore-microtubule attachments. Aurora B has been shown to phosphorylate and regulate MCAK both in vitro and in vivo (167). Aurora B activity was essential for the localization of MCAK to centromeres, but not to spindle poles. Upon Aurora B-induced phosphorylation of serine 196 in the neck region of MCAK, microtubule depolymerization activity was inhibited (167). Demonstrating a direct link between the microtubule depolymerase MCAK and Aurora B kinase.

SUMMARY

The organization of interphase microtubules and mitotic spindles are dramatically altered during the cell cycle and development and this organization is highly dependent on interactions between microtubules and accessory proteins. The highly coordinated events that lead to separation and segregation of chromosomes are highly dependent of microtubule dynamics. The molecular mechanisms underlying this dynamic behavior are multifaceted and not yet fully understood. What is clear however is that microtubule

regulating factors such as MAPs and other microtubule interacting proteins that either reduce or enhance microtubule dynamics during different phases of the cell cycle, are essential for many cellular functions.

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